

Binary Compounds of Rhodium and Zinc: RhZn, Rh₂Zn₁₁, and RhZn₁₃

Nicole Gross, Gunter Kotzyba, Bernd Künnen, and Wolfgang Jeitschko*

Münster, Anorganisch-Chemisches Institut der Westfälischen Wilhelms-Universität

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Abstract. The title compounds were prepared by reaction of the elements at elevated temperatures in sealed silica tubes. Single crystals of RhZn and RhZn₁₃ were obtained by slow cooling of samples with a high zinc content after dissolving the zinc-rich matrix in hydrochloric acid. Their crystal structures were determined from single-crystal X-ray diffractometer data. RhZn has a CsCl type structure: $Pm\bar{3}m$, $a = 300.9(1)$ pm. RhZn₁₃ has a CoZn₁₃ type structure: $C2/m$, $a = 1090.8(2)$ pm, $b = 753.7(2)$ pm, $c = 512.7(1)$ pm, $\beta = 101.02(2)^\circ$. The structure of Rh₂Zn₁₁ is isotypic with Cu₅Zn₈, the γ -brass structure. It was refined from X-ray diffractometer powder data: $I\bar{4}3m$, $a = 909.1(1)$ pm. In these structures all atoms have high coordination numbers. The struc-

ture of RhZn₁₃ contains relatively large unoccupied voids. It is suggested that they accommodate nonbonding electrons. Electrical conductivity measurements of Rh₂Zn₁₁ and RhZn₁₃ indicate metallic behavior, however, with an unexpectedly high resistivity for Rh₂Zn₁₁. The expected Pauli paramagnetism of these compounds is overcompensated by the core diamagnetism, suggesting a low density of states at the Fermi level especially for Rh₂Zn₁₁. This correlates with the high electrical resistivity of this compound.

Keywords: Rhodium; Zinc; Crystal structure; Electrical conductivity; Pauli paramagnetism

Binäre Verbindungen von Rhodium und Zink: RhZn, Rh₂Zn₁₁ und RhZn₁₃

Inhaltsübersicht. Die Titelverbindungen wurden durch Reaktion der Elemente bei hohen Temperaturen in evakuierten Quarzglasampullen hergestellt. Einkristalle von RhZn und RhZn₁₃ wurden durch langsames Abkühlen von Proben mit hohem Zinkgehalt nach Auflösen der zinkreichen Matrix in Salzsäure erhalten. Ihre Kristallstrukturen wurden anhand von Einkristall-Diffraktometerdaten bestimmt. RhZn hat CsCl-Struktur: $Pm\bar{3}m$, $a = 300.9(1)$ pm. RhZn₁₃ kristallisiert in der CoZn₁₃-Struktur: $C2/m$, $a = 1090.8(2)$ pm, $b = 753.7(2)$ pm, $c = 512.7(1)$ pm, $\beta = 101.02(2)^\circ$. Die Struktur von Rh₂Zn₁₁ ist isotyp mit Cu₅Zn₈ (γ -Messing). Sie wurde anhand von Röntgen-Pulverdiffraktometerdaten verfeinert: $I\bar{4}3m$, $a = 909.1(1)$ pm. In diesen Strukturen haben alle Ato-

me hohe Koordinationszahlen. In der Struktur von RhZn₁₃ treten relativ große unbesetzte Lücken auf. Es wird vermutet, dass sie nichtbindende Elektronen beinhalten. Elektrische Leitfähigkeitsmessungen von Rh₂Zn₁₁ und RhZn₁₃ zeigen metallisches Verhalten, jedoch mit einem ungewöhnlich hohen elektrischen Widerstand von Rh₂Zn₁₁. Der zu erwartende Pauli-Paramagnetismus dieser Verbindungen wird durch den Diamagnetismus der Rumpfelektronen überkompensiert. Besonders ausgeprägt ist dies für Rh₂Zn₁₁, was eine geringe Zustandsdichte an der Fermi-Kante erwarten läßt. Dies korreliert mit dem hohen elektrischen Widerstand dieser Verbindung.

Introduction

Apparently, the binary system of rhodium with zinc has never been investigated. Neither Moffatt's compilation [1] with subsequent updates nor the handbook by Massalski [2] provide any information on that system. Also, the most recent editions of Pearson's Handbook [3, 4] do not list any binary compounds of rhodium with zinc. We have also searched without success for such compounds in *Chemical Abstracts*. In the course of our investigations on compounds with high zinc content, e.g., Ti₃Zn₂₂ and TiZn₁₆ [5], Zr₅Zn₃₉ and ZrZn₂₂ [6], NbZn₃, NbZn₁₆, and MoZn_{20.44} [7] we also investigated binary compounds

of rhodium with zinc. The results are reported here. A brief account of this work has already been given at a conference [8].

Sample Preparation and Lattice Constants

Starting materials for the preparation of the samples were rhodium powder (Degussa) and zinc ingots (Merck), all with stated purities greater than 99.9%. The three compounds RhZn, Rh₂Zn₁₁, and RhZn₁₃ were isolated from samples with the atomic ratios Rh:Zn = 1:2, 1:10, and 1:20, which were sealed in evacuated silica tubes. They were heated to 850 °C at a rate of 50 °C/h, held at that temperature for 8 days, and subsequently cooled to room temperature at a rate of 2 °C/h. It must be mentioned that unreacted zinc accumulates at the cooler parts of the sample tubes. For that reason the samples were shaken every 5 to 10 hours during the annealing. After breaking the sample tubes most of the excess zinc could be separated from the samples mechanically. Nevertheless, the crushed ingots were treated in di-

* Prof. Dr. W. Jeitschko
Anorganisch-Chemisches Institut
Wilhelm-Klemm-Strasse 8
D-48149 Münster, Germany

luted hydrochloric acid to remove any remaining zinc. The binary rhodium-zinc compounds were practically not attacked by this treatment. They were obtained in the form of well-developed crystals. Energy-dispersive X-ray fluorescence analyses in a scanning electron microscope did not reveal any impurity elements heavier than sodium.

All three compounds were also prepared from powders of the elements with the ideal atomic ratios by the same annealing conditions as described above. These samples were not treated by hydrochloric acid. They were used for the electrical conductivity and magnetic susceptibility measurements.

Guinier powder patterns of the compounds were recorded with $\text{CuK}\alpha_1$ radiation using α -quartz ($a = 491.30$ pm, $c = 540.46$ pm) as an internal standard. The following lattice constants were obtained by least-squares fits of these data. RhZn : cubic, $a = 300.9(1)$ pm, $V = 0.02724(3)$ nm³, $\text{Rh}_2\text{Zn}_{11}$: cubic, $a = 909.1(1)$ pm, $V = 0.7513(3)$ nm³, and RhZn_{13} : monoclinic, $a = 1090.8(2)$ pm, $b = 753.7(2)$ pm, $c = 512.7(1)$ pm, $\beta = 101.02(2)^\circ$, $V = 0.4137(3)$ nm³.

Properties

The well crystallized compounds RhZn , $\text{Rh}_2\text{Zn}_{11}$, and RhZn_{13} are stable on air for long periods of time. They are silvery with metallic luster.

The electrical conductivities of $\text{Rh}_2\text{Zn}_{11}$ and RhZn_{13} were determined with a four-probe technique in the temperature range from 4 K to room temperature. The electrical contacts were made by fine copper wires, which were attached to the samples with well conducting silver epoxy pastes. The equiatomic compound RhZn was obtained only in the form of porous ingots or in the form of small crystals, and the cold-pressed pellets were too brittle and therefore unsuited for electrical conductivity measurements.

The electrical conductivity of $\text{Rh}_2\text{Zn}_{11}$ was determined for several compact polycrystalline samples. As could be expected, the compound shows metallic behavior with increasing resistivity for increasing temperature (Figure 1). However, the absolute values of the electrical resistivities are rather high, a factor of 200–400 higher than the resistivities of compounds with similar compositions, e. g., $\text{Ti}_3\text{Zn}_{22}$ [5], TiZn_{16} [5], and $\text{Zr}_5\text{Zn}_{39}$ [6]. Nevertheless, these results were reproduced twice with different samples. For the nearly linear dependence of the electrical resistivity with temperature (T) in the range from 200 to 300 K the resistivity of $\text{Rh}_2\text{Zn}_{11}$ can be described by the equation $\rho(T) = 3694(4) \mu\Omega\text{cm} + T \cdot 1.70(2) \mu\Omega\text{cm} \cdot \text{K}^{-1}$. The figures given in parentheses correspond to standard deviations in the place values of the last listed digits. The equation describes the relative resistivities for different temperatures. The absolute values of the resistivities are estimated to be correct within $\pm 25\%$.

In contrast, the electrical resistivity determined by us for RhZn_{13} shows the expected metallic behavior

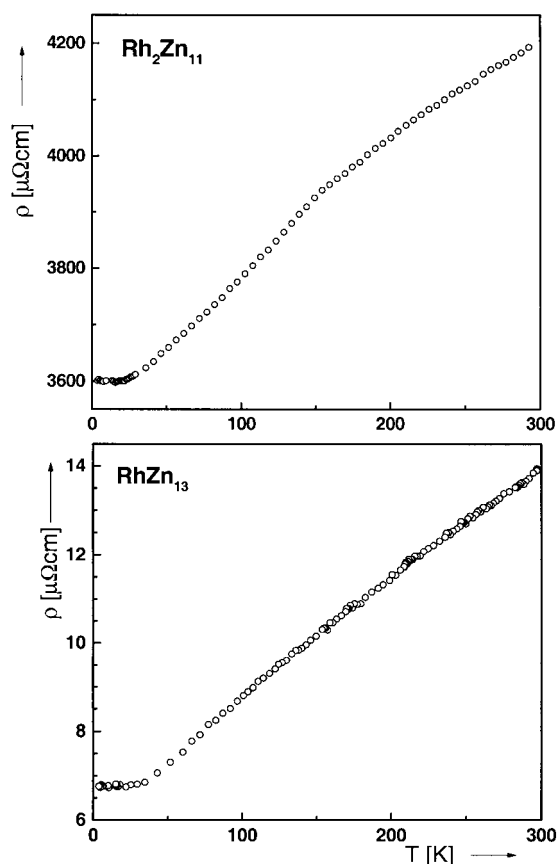


Fig. 1 Electrical resistivities of $\text{Rh}_2\text{Zn}_{11}$ and RhZn_{13} as a function of temperature.

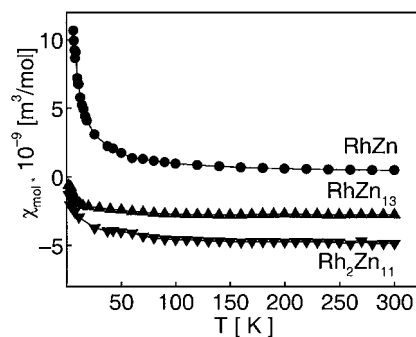


Fig. 2 Temperature dependence of the magnetic susceptibilities of RhZn , $\text{Rh}_2\text{Zn}_{11}$, and RhZn_{13} determined with a magnetic flux density of one Tesla.

with low values for the specific resistivities (Figure 1). Again, these are estimated to be correct only within $\pm 25\%$, whereas the relative values for different temperatures are much more accurate. The least-squares fit of these data for the linear behavior of RhZn_{13} in the temperature range between 150 and 300 K resulted in the equation $\rho(T) = 6.49(3) \mu\Omega\text{cm} + T \cdot 0.0249(1) \mu\Omega\text{cm} \cdot \text{K}^{-1}$. The absolute value of the electrical resistivity of RhZn_{13} at 273 K amounts to $\rho = 13 \pm 3 \mu\Omega\text{cm}$. It is somewhat higher than the re-

sistivities of the elemental components at 273 K: $\rho(\text{Rh}) = 4.35 \mu\Omega\text{cm}$ [9] and $\rho(\text{Zn}) = 5.65 \mu\Omega\text{cm}$ [10].

The magnetic susceptibilities of polycrystalline samples of RhZn, Rh₂Zn₁₁, and RhZn₁₃ were determined with a SQUID magnetometer (Figure 2). The susceptibility values are small and down to 50 K nearly temperature independent, indicating Pauli paramagnetism. These values were fitted to the modified Curie-Weiss law $\chi = \chi_0 + C(T - \Theta)^{-1}$. For the three compounds RhZn, Rh₂Zn₁₁, and RhZn₁₃ the temperature independent susceptibility values χ_0 amounted to $0.25(2) \cdot 10^{-9} \text{ m}^3/\text{formula unit}$, $-4.9(1) \cdot 10^{-9} \text{ m}^3/\text{f.u.}$, and $-2.76(2) \cdot 10^{-9} \text{ m}^3/\text{f.u.}$, respectively. The negative values for Rh₂Zn₁₁ and RhZn₁₃ indicate that the Pauli paramagnetism is overcompensated by the core diamagnetism of these compounds. Below 50 K the magnetic susceptibilities increase. The magnetic moments μ_{exp} obtained from the least-squares fits of the temperature dependent parts of the susceptibilities of the modified Curie-Weiss law are small: $0.21(2) \mu_{\text{B}}$, $0.14(2) \mu_{\text{B}}$, and $0.03(1) \mu_{\text{B}}$ for RhZn, Rh₂Zn₁₁, and RhZn₁₃, respectively. Such upturns of the magnetic susceptibilities at low temperatures (“Curie tails”) are frequently observed for Pauli paramagnetic samples. They can be ascribed to paramagnetic impurities and/or paramagnetic surface states, even though the Guinier powder patterns did not reveal any impurities in the samples.

Structure Refinements

Single crystals of RhZn and RhZn₁₃ were isolated from the annealed samples and selected for the X-ray data collections on the basis of their *Laue* patterns. The intensity data were recorded on a four-circle diffractometer (Enraf-Nonius, CAD4) with graphite-monochromated MoK α radiation and a scintillation counter with pulse-height discrimination. The scans were along the *Laue* streaks ($\theta/2\theta$) with background counts at both ends of each scan. Empirical absorption corrections were made on the basis of psi-scans. Further details concerning the data collections are listed in Table 1.

The structures of RhZn and RhZn₁₃ were determined and refined with the program system SHELX-97 [11]. The single-crystal of RhZn showed the primitive cubic cell of the CsCl type structure. For RhZn₁₃ a *C* centered monoclinic cell was established. Eventually RhZn₁₃ was recognized to be isotypic with CoZn₁₃ [12].

For the structure refinements a full-matrix least-squares program was used with atomic scattering factors corrected for anomalous dispersion as provided by the program. The weighting schemes accounted for the counting statistics, and parameters correcting for isotropic secondary extinction were optimized as least-squares variables.

To check for deviations from the ideal compositions, occupancy parameters were refined together with variable thermal parameters in separate series of least-squares cycles. For RhZn the light (Zn) position was found to have slightly

Table 1 Crystallographic data of RhZn and RhZn₁₃

Compound	RhZn	RhZn ₁₃
Formula mass	168.29	952.8
Space group	<i>Pm</i> $\bar{3}$ <i>m</i> (No. 221)	<i>C2/m</i> (No. 12)
Lattice constants (pm)	<i>a</i> = 299.97(6)	<i>a</i> = 1090.8(4) <i>b</i> = 753.7(2) <i>c</i> = 512.7(1) β = 101.02(3)°
(four-circle diffractometer)		
Formula units/cell, <i>Z</i>	1	2
Pearson symbol	cP2	mC28
Calculated density (g/cm ³)	10.26	7.65
Crystal size (μm ³)	10 × 10 × 20	80 × 20 × 10
Max/min transmission	1.07	1.05
2 θ range up to (°)	68	64
Range in <i>hkl</i>	±4, ±4, 0–4	±16, 0–11, ±7
Total no. of reflections	264	1674
Independent reflections	23	766
<i>R</i> _{int} (<i>F</i> ²)	0.093	0.077
Reflections with <i>I</i> _o > 2σ(<i>I</i> _o)	23	385
Number of variables	5	39
<i>R</i> (No. of <i>F</i>)	0.025 (23)	0.038 (385)
<i>R</i> _w (No. of <i>F</i> ²)	0.047 (23)	0.112 (766)
Highest residual peaks (e/Å ³)	0.87	1.88

more scattering power than what would be expected for the ideal composition. This allows two different extreme refinements. One with ideal occupancy of the rhodium position and mixed Zn/Rh occupancy of the other, lighter position, and the other extreme with ideal occupancy of the zinc position and with mixed Rh/Zn occupancy of the other (heavier) position. The resulting compositions obtained for this crystal, not allowing for vacant sites, were Rh_{1.017(3)}Zn_{0.983} and Rh_{0.976(2)}Zn_{1.024}. Mathematically it is not possible to distinguish between these two extreme cases, and allowing for mixed occupancy of both positions the ideal composition is possible. Since our crystal of RhZn was obtained from a sample with the overall composition Rh/Zn = 1 : 2, a rhodium content smaller than 50% could be expected. Our structure refinement for this case shows that the rhodium/zinc ratio can at most be as small as 0.976/1.024 = 0.953. Since, for thermodynamic reasons, a mixed occupancy for *both* positions can be expected, this ratio must be greater in reality. If vacant sites are allowed for the rhodium position, and the zinc site has the ideal occupancy, a composition of Rh_{0.992(1)}Zn is obtained, i. e., 0.8% of the rhodium sites would not be occupied. For the occupancy parameters of RhZn₁₃ the values varied between 99.9(2)% for the Zn3 site and 100.6(2)% for the Zn4 site. Hence, for both compounds no great deviations from the ideal compositions were found, and in the final refinement cycles we assumed the ideal occupancy values for both compounds.

The final difference Fourier analyses showed no significant electron densities at sites suitable for additional atomic positions. The final residuals, atomic parameters, and the interatomic distances are given in the Tables 1–3.

The structure of Rh₂Zn₁₁ was recognized to be isotypic with γ -brass Cu₅Zn₈ [14–16] and Ir₂Zn₁₁ [17] from its cubic powder pattern and from the estimated cell content by comparison with the compilation in Pearson's Handbook [3, 4]. Since we were not successful in obtaining single crystals of this compound, we refined this structure from X-ray powder diffractometry.

Table 2 Atomic parameters of RhZn, Rh₂Zn₁₁, and RhZn₁₃^{a)}

Atom	Site	x	y	z	U ₁₁	U ₂₂	U ₃₃	U ₁₂	U ₁₃	U ₂₃	U (pm ²)
RhZn (<i>Pm3m</i>)											
Rh	1 b	0	0	0	58(1)	U ₁₁	U ₁₁	0	0	0	58(1)
Zn	1 a	1/2	1/2	1/2	49(1)	U ₁₁	U ₁₁	0	0	0	49(1)
Rh₂Zn₁₁ (<i>I43m</i>)											
Rh	8 c	0.3268(3)	0.3268(3)	0.3268(3)	–	–	–	–	–	–	215(12)
Zn1	24 g	0.3119(3)	0.3119(3)	0.0368(3)	–	–	–	–	–	–	156(7)
Zn2	12 e	0.3564(6)	0	0	–	–	–	–	–	–	156(7)
Zn3	8 c	0.1099(4)	0.1099(4)	0.1099(4)	–	–	–	–	–	–	156(7)
RhZn₁₃ (<i>C2/m</i>)											
Rh	2 c	0	0	1/2	40(2)	47(2)	37(2)	0	14(2)	0	41(1)
Zn1	8 j	0.07905(4)	0.30257(7)	0.3236(1)	140(2)	159(2)	230(2)	–15(2)	20(2)	114(2)	178(1)
Zn2	8 j	0.32145(4)	0.32036(6)	0.1977(1)	98(2)	111(2)	123(2)	–15(2)	–21(2)	–31(2)	116(1)
Zn3	4 i	0.22174(5)	0	0.3645(1)	99(2)	194(3)	157(3)	0	82(2)	0	142(2)
Zn4	4 i	0.38741(5)	0	0.0701(1)	82(2)	77(2)	134(2)	0	63(2)	0	92(1)
Zn5	2 a	0	0	0	157(4)	382(6)	50(3)	0	48(3)	0	193(3)

^{a)} The positional parameters were standardized with the program STRUCTURE TIDY [13]. The anisotropic displacement parameters are defined by $T = \exp[-2\pi^2(h^2a^{*2}U_{11} + \dots + 2klb^*c^*U_{23})]$. The last column contains the isotropic displacement parameters of RhZn and Rh₂Zn₁₁ and the equivalent isotropic displacement parameters of the atomic positions of RhZn₁₃.

Table 3 Interatomic distances in RhZn, Rh₂Zn₁₁, and RhZn₁₃^{a)} in pm

RhZn					
Rh:	8 Zn	260.6	Zn:	8 Rh	260.6
	6 Rh	300.9		6 Zn	300.9
Rh₂Zn₁₁					
Rh:	3 Zn1	261.2	Zn2:	2 Zn1	260.6
	3 Zn1	264.3		1 Zn2	261.1
	3 Zn3	269.9		2 Zn3	264.9
	3 Zn2	278.1		2 Rh	278.1
	(1 Zn3	341.5)		4 Zn1	288.4
Zn1:	1 Zn2	260.6		2 Zn1	292.2
	1 Rh	261.2	Zn3:	3 Zn2	264.9
	1 Rh	264.3		3 Zn1	268.1
	1 Zn3	268.1		3 Rh	269.9
	4 Zn1	270.9		3 Zn3	282.6
	2 Zn2	288.4		(1 Rh	341.5)
	1 Zn2	292.2			
	(2 Zn1	348.5)			
	(2 Zn1	353.7)			
RhZn₁₃					
Rh:	2 Zn5	256.4		1 Zn4	277.7
	4 Zn2	262.2		1 Zn1	284.2
	2 Zn3	264.1		1 Zn3	284.6
	4 Zn1	265.7		1 Zn1	291.2
Zn1:	1 Zn4	258.9		1 Zn3	313.7
	1 Zn2	265.4		1 Zn1	320.8
	1 Rh	265.7	Zn3:	1 Zn4	256.6
	1 Zn4	269.0		1 Rh	264.1
	1 Zn1	272.7		2 Zn2	273.8
	1 Zn3	274.6		2 Zn1	274.6
	1 Zn2	284.2		1 Zn5	275.9
	1 Zn5	285.5		2 Zn2	284.6
	1 Zn3	285.7		2 Zn1	285.7
	1 Zn2	291.2		2 Zn2	313.7
	1 Zn1	297.6	Zn4:	1 Zn3	256.6
	1 Zn2	320.8		2 Zn1	258.9
	1 Zn4	340.2		2 Zn2	263.7
	1 Zn1	343.5		1 Zn4	268.8
Zn2:	1 Zn2	254.1		2 Zn1	269.0
	1 Rh	262.2		2 Zn2	277.7
	1 Zn4	263.7		2 Zn1	340.2
	1 Zn1	265.4	Zn5:	2 Rh	256.4
	1 Zn2	270.8		4 Zn2	272.2
	1 Zn5	272.2		2 Zn3	275.9
	1 Zn3	273.8		4 Zn1	285.5

^{a)} The distances were calculated with the lattice constants as obtained from the Guinier powder data. All distances shorter than 370 pm are listed. Standard deviations are all equal or less than 0.1 pm in RhZn and RhZn₁₃. In Rh₂Zn₁₁ they are all equal or less than 0.8 pm. Near neighbors with distances listed in parentheses are not shown in the coordination polyhedra of Figs. 4 and 5.

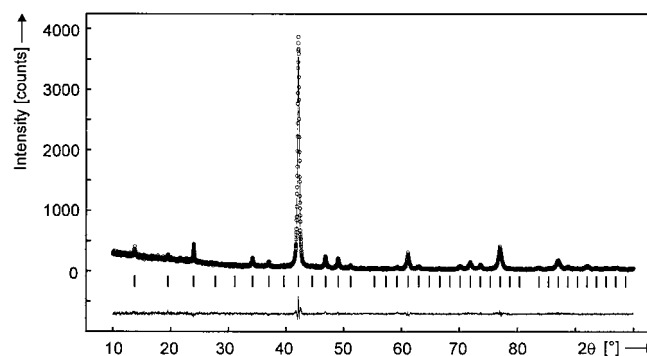


Fig. 3 X-ray powder diffractogram of Rh₂Zn₁₁ with Cu₅Zn₈ type structure. The observed profile (dots), the calculated profile (continuous line), the line positions, and the difference spectrum are shown.

meter data with the Rietveld method using the program FULLPROF [18]. This refinement confirmed the Cu₅Zn₈ type structure: space group *I43m*, No. 217, Z = 4, Pearson symbol cI52, formula mass 924.99, calculated density $\rho_c = 8.18 \text{ g/cm}^3$.

The X-ray intensity data of Rh₂Zn₁₁ (Figure 3) were recorded from a fine powder, placed between acetate foils. A rotating sample holder was used on a focussing powder diffractometer (STOE Stadi P) with monochromated CuK α_1 radiation and a linear position-sensitive detector in transmission geometry. A total of 4501 intensity data was collected in the 2-theta range 10–100° with a step width of 0.02°.

The atomic positions of Ir₂Zn₁₁ were used as starting parameters. The atomic scattering factors, corrected for anomalous dispersion, were provided by the least-squares program [18]. In the final refinement cycles a total of 12 parameters was optimized, including five positional parameters, two thermal parameters, the zero point, the scale factor, the lattice constant of Rh₂Zn₁₁, and two parameters to fit the peak profile with a pseudo-Voigt function. The number of reflections resulting from this compound was 54. The positional parameters could be refined independently,

while the isotropic displacement parameters of the zinc atoms were set to be equal. The lattice constant of this sample, obtained from the powder diffractometer data, was 908.79(2) pm, in good agreement with the lattice constant of 909.1(1) pm from the Guinier powder data. The goodness of fit (χ^2) was 1.38, while the residuals amounted to $R_{\text{Bragg}} = 0.059$ and $R_F = 0.076$. The atomic parameters are listed in Table 2 and the interatomic distances in Table 3.

Discussion

The rhodium-zinc compounds characterized here are the first ones reported of that binary system. However, in related transition metal-zinc systems, compounds with similar structures and compositions are known. RhZn has CsCl type structure, which has also been observed for NiZn [19], $\sim$ Pd₂Zn₃ [20], $\sim$ Pd₂Cd₃ [21], CuZn [22–24], AgZn [25], and AuZn [26–28]. Rh₂Zn₁₁

is isotypic with Cu₅Zn₈ [14–16] and Ir₂Zn₁₁ [17]. This structure has also been found for $\sim$ Fe₃Zn₁₀ [16, 29, 30], $\sim$ Ni₂Zn₁₁ [29], $\sim$ Pd₂Zn₁₁ [31], Cu₅Cd₈ [15], Ag₅Zn₈ [32, 33], Ag₅Cd₈ [34], and Au₅Cd₈ [35]. RhZn₁₃ has CoZn₁₃ type structure [12]. This structure has only two more representatives [3, 4]: MnZn₁₃ [12] and FeZn₁₃ [12].

All of the above mentioned compounds are in the regime, where it is generally agreed that electron counting rules are important. Usually the electron per atom ratio e/a is considered. The best known examples are β -CuZn (*bcc*) with an e/a ratio of $3/2 = 21/14$, Cu₅Zn₈ (γ -brass) with $e/a = 21/13$, and ε -CuZn₃ (*hcp*) with $e/a = 7/4 = 21/12$ [36, 37]. Compounds with similar electron counts are isotypic, e. g., Ag₃Al with $e/a = 21/14$ has a body centered cubic structure, Cu₉Al₄ with $e/a = 21/13$ is isotypic with Cu₅Zn₈ and Ag₅Cd₈, and Ag₅Al₃ with $e/a = 21/12$ has a hexagonal close packed structure [3, 15, 38, 39]. For this counting rule one

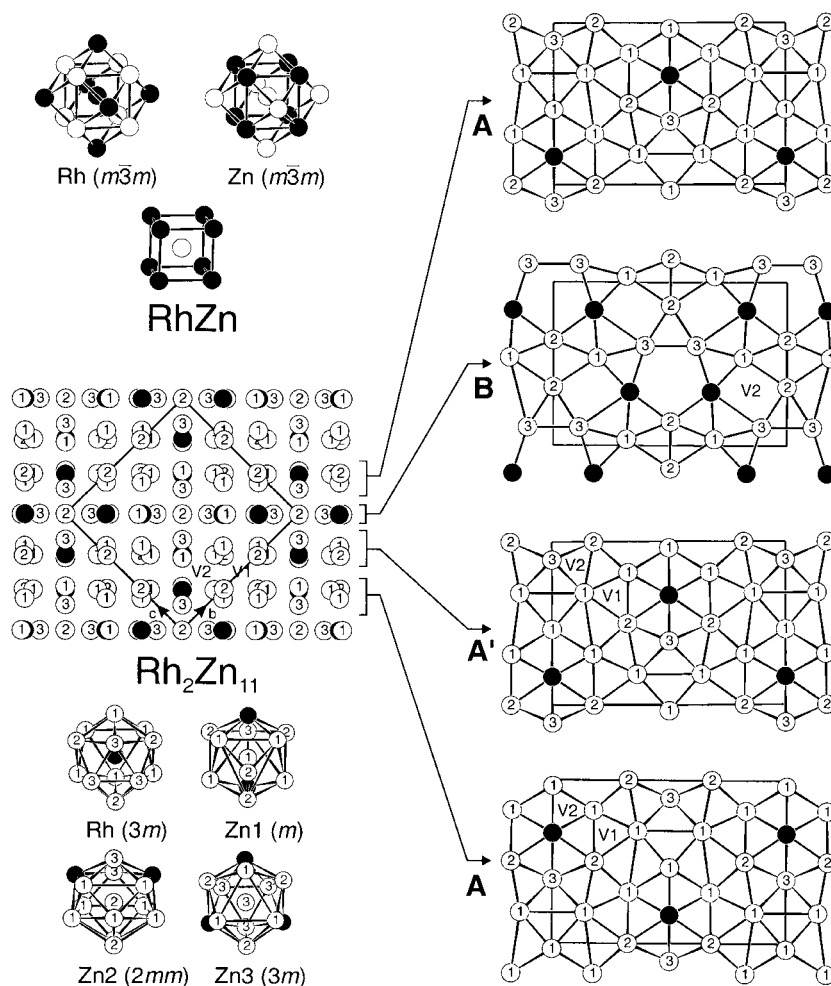


Fig. 4 Crystal structure and coordination polyhedra of RhZn (CsCl type) and Rh₂Zn₁₁ (Cu₅Zn₈ type). The structure of Rh₂Zn₁₁ may be visualized as consisting of atomic layers of two kinds, designated with the letters **A** and **B**, which are shown on the right-hand side. The atoms on the layer **B** are situated on a mirror plane. Single-digit numbers correspond to the atom designations. The site symmetries of the central atoms of the coordination polyhedra are given in parentheses. Also shown are the positions of two different octahedral interstitial sites (voids V1 and V2).

electron is counted for Cu, Ag, and Au, two electrons for Zn and Cd, and three for Al, Ga, or In. Similar results are obtained when the total number of *s*, *p*, and *d* valence electrons is counted, i. e., for the elements Fe, Co, Ni, Cu, Zn, and Ga one counts 8, 9, 10, 11, 12, and 13 electrons (the modern group number), respectively [40]. For instance, we obtain for the *bcc* compounds CuZn and Ag₃Al electron counts of $(11 + 12)/2 = 11.50$ and $(33 + 13)/4 = 11.50$, respectively; for the γ -brass type compounds Cu₅Zn₈, Cu₉Al₄, and Ni₂Zn₁₁ we obtain electron counts of $(55 + 96)/13 = 11.62$, $(99 + 52)/13 = 11.62$, and $(20 + 132)/13 = 11.69$, respectively; and finally, for the hexagonal close packed (*hcp*) structure of CuZn₃ and Ag₅Al₃ the electron

counts amount to $(11 + 36)/4 = 11.75$ and $(55 + 39)/8 = 11.75$. For the new compound RhZn with (ordered) *bcc* type structure we obtain an electron count of $(9 + 12)/2 = 10.50$ which is lower than the count of 11.50 for CuZn and Ag₃Al. A better agreement is obtained for Rh₂Zn₁₁ with an electron count of $(18 + 132)/13 = 11.54$, as compared to the electron counts of 11.62, 11.62, and 11.69 for Cu₅Zn₈, Cu₉Al₄, and Ni₂Zn₁₁, respectively.

The crystal structures and the coordination polyhedra of the three compounds RhZn, Rh₂Zn₁₁, and RhZn₁₃ are shown in Figures 4 and 5. As is usually the case for intermetallic compounds, all atoms have high coordination numbers. For the CsCl type struc-

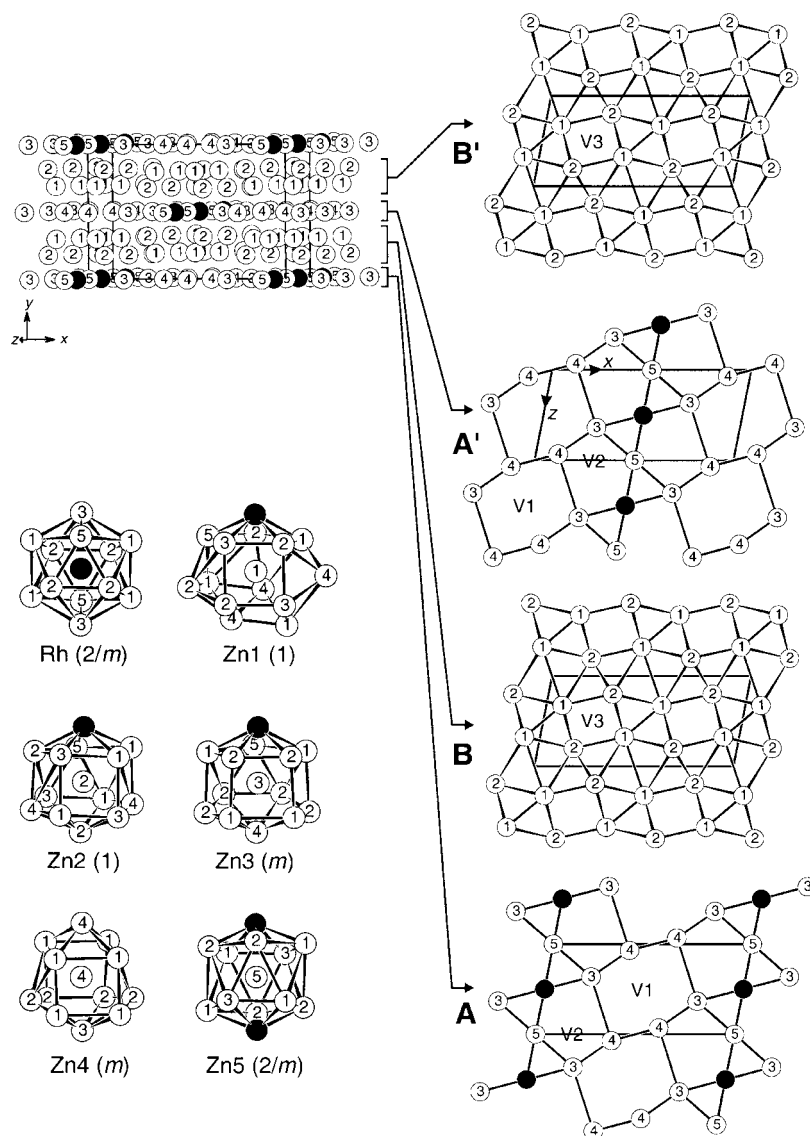


Fig. 5 Crystal structure and coordination polyhedra of RhZn₁₃. The monoclinic structure is shown in a projection along a direction perpendicular to the *ab* plane in the upper left-hand corner. The structure may be considered as a stacking of the two layers **A** and **B**, which are viewed along the *b* axis on the right-hand side. The layers **A** are situated on the mirror plane of the *C2/m* cell. The layers **B** are puckered. The positions of three relatively large voids V1, V2, and V3 are indicated.

ture of RhZn we consider both kinds of atoms to have the coordination number (CN) 14: 8 near neighbors of the other kind plus 6 neighbors of the same kind. The 6 neighbors of the same kind (at 300.9 pm) are only 15% further away than the 8 neighbors of the other kind (260.6 pm). These distances may be compared to the sums of the metallic radii for CN 12, as calculated by us from the structures of the elements [41], which amount to 269.0, 273.4, and 277.8 pm for the Rh–Rh, Rh–Zn, and Zn–Zn interactions, respectively. The weighted average distance for the 14 near neighbors of the Rh and Zn atoms is 277.9 pm, and considering that this distance corresponds to a larger CN, this distance agrees very well with the sum of the metallic radii. Thus, for the intermetallic compound RhZn we consider all three kinds of atomic interactions (Rh–Rh, Rh–Zn, and Zn–Zn) as bonding. This is certainly in contrast to an ionic compound, e.g., CsCl, where CN 8 is appropriate, since the Cs–Cs and Cl–Cl interactions are considered as antibonding. It seems likely that at high temperature the structure of RhZn is disordered as it is well known for CuZn. We have not searched for this suspected order-disorder transition.

The structure of Rh₂Zn₁₁ contains only four atomic sites. The rhodium site has an environment of 12 zinc atoms forming a slightly distorted icosahedron. The average Rh–Zn distance amounts to 268.4 pm, slightly less than the sum of the metallic radii of 273.4 pm (again for CN 12). The small difference may be taken as an indication for the Rh–Zn interactions in Rh₂Zn₁₁ to be stronger than the Rh–Rh and Zn–Zn interactions in the elements. The Zn1, Zn2, and Zn3 atoms have 11, 13, and 12 neighbors, respectively, of which 2 (Zn1 and Zn2) or 3 (Zn3) are rhodium atoms. The weighted average Zn–Zn distance amounts to 275.4 pm. It is slightly less than the average Zn–Zn distance of 277.8 pm in elemental zinc. Qualitatively, this agrees with the fact that the weighted average CN of the zinc atoms with 11.7 is slightly less than 12. However, the structure of Rh₂Zn₁₁ was only refined from powder data, and one should not overinterpret such small differences in interatomic distances.

As already mentioned above, the structure of Rh₂Zn₁₁ is isotypic with Cu₅Zn₈ in the sense that it has the same atomic positions as Cu₅Zn₈ [14–16], the γ -brass type structure. An atomic distribution corresponding to Rh₂Zn₁₁ has already been observed for Ir₂Zn₁₁ [17]. In Figure 4 we have somewhat arbitrarily dismantled this cubic structure in two kinds (**A** and **B**) of more or less close packed atomic layers, to facilitate the visualization of this rather complex atomic arrangement. The γ -brass type structure has sometimes been counted to belong to the family of tetrahedrally close packed structures, which contain only tetrahedral interstitial sites [16, 30, 39]. This is somewhat surprising, since it is well known that tetrahedral close

packing is only possible for spheres of unequal size, e.g., in the σ -phase related structures and in the Laves phases [42]. Since the rhodium and zinc atoms are of approximately the same size, tetrahedral close packing is unexpected for Rh₂Zn₁₁. We therefore looked for interstitial sites in the structure of Rh₂Zn₁₁, and indeed there are at least two interstitial sites in this structure with more or less distorted octahedral coordination. These voids (V) and their atomic environments are listed in Table 4. They are large enough to accommodate additional small (interstitial) atoms, e.g., carbon or oxygen. However, we have no reason to believe that these voids are occupied to any large extent by such atoms in Rh₂Zn₁₁ or any other of the many intermetallic compounds with γ -brass type structure. Nevertheless, the γ -brass type structure should not be considered as tetrahedrally close packed.

The structure of RhZn₁₃ has only six different atomic sites. The Rh site has almost perfect icosahedral zinc coordination with Rh–Zn distances varying between 256.4 and 265.7 pm and a weighted average distance of 262.7 pm. The latter is considerably smaller than the average Rh–Zn distance of 268.4 pm in the RhZn₁₂ icosahedron of Rh₂Zn₁₁: a difference of 5.7 pm. Relatively short interatomic distances are sometimes observed between partially occupied sites. However, the structure refinement of RhZn₁₃ showed practically full occupancy (100.0 ± 0.6%) for all atomic sites. In part, this difference can be rationalized by the larger spread of Rh–Zn distances in Rh₂Zn₁₁ (Table 3), and in part also by the fact that the Rh atom in Rh₂Zn₁₁ has a thirteenth neighbor at 341.5 pm (In RhZn₁₃ the next zinc atom outside the RhZn₁₂ icosahedron is at 442.0 pm). Nevertheless, even in view of these well known effects we find this difference remarkable.

The five different zinc sites of RhZn₁₃ have CN's between 12 and 14 (Figure 5). The Zn4 atom has no rhodium neighbor and the Zn5 atom has two rhodium neighbors; the others have one. The Zn–Zn distances almost continuously cover the range between 254.1 and 343.5 pm. The average Zn–Zn distances amount to 291.5, 281.0, 283.1, 278.7, and 278.3 pm, respectively, for the sequence from Zn1 to Zn5. All of these distances are larger than the overall average Zn–Zn distance of 275.3 pm in Rh₂Zn₁₁. This can easily be rationalized by the fact that the CN's of the zinc atoms in RhZn₁₃ are larger: The weighted average CN of the zinc atoms in RhZn₁₃ is 13.1 as compared to 11.7 in Rh₂Zn₁₁.

The structure of RhZn₁₃ can be visualized as consisting of atomic layers, although strong chemical bonds extend in all three dimensions as can readily be concluded by looking at the coordination polyhedra. The densely packed layers of the type **B** (Figure 5) contain only zinc atoms, while the less densely packed layers of the type **A** contain both, rhodium and zinc

Table 4 Location and coordination (pm) of unoccupied sites (voids V) in the structures of Rh₂Zn₁₁ and RhZn₁₃

Rh ₂ Zn ₁₁	$\bar{1}\bar{4}3m$	x	y	z
V1	24 f	0.637	1/2	0
V2	24 g	0.170	0.17	0.34
V1: 2 Zn1	180.3		V2: 2 Zn1	178.8
2 Zn2	180.4		1 Zn1	180.4
2 Zn1	235.9		1 Rh	202.0
			1 Zn2	219.1
			1 Zn3	223.0

RnZn ₁₃	$C2/m$	x	y	z	
V1	2 d	1/2	0	1/2	
V2	4 i	0.2225	0	0.0210	
V3	4 f	1/4	1/4	1/2	
V1: 4 Zn1	201.9	V2: 1 Zn3	176.3	V3: 2 Zn2	193.8
2 Zn4	230.9	2 Zn2	176.5	2 Zn1	195.1
2 Zn3	298.2	1 Zn4	176.8	2 Zn3	201.2
		1 Zn5	240.9		
		2 Zn2	272.8		

atoms. Again, as in Rh₂Zn₁₁, we can discern relatively large voids in the structure of RhZn₁₃ which are marked V1, V2, and V3 in Figure 5. Their positions and atomic environments are listed in Table 4. These voids are even larger than in Rh₂Zn₁₁. As was discussed earlier for the structures of Ti₃Zn₂₂ [5], TiZn₁₆ [5], Zr₅Zn₃₉ [6], ZrZn₂₂ [6], and MoZn_{20.44} [7], we believe these voids to accommodate nonbonding or only weakly bonding electrons, as this is well known for the stereochemically active lone pairs of the main group elements in their lower states, e.g., for Sn²⁺, Pb²⁺, Sb³⁺, Bi³⁺, Te⁴⁺, etc. This tendency is already indicated by the large c/a ratio of the hexagonally “close packed” structure of elemental zinc. The most convincing structure in this respect is that of Zr₅Zn₃₉, a defect variant of the structure found for the compounds Ln₅Mg₄₁ (Ln = Ce–Sm). Apparently, these two structures have an optimal electron count which can be achieved in the structure of Zr₅Zn₃₉ only by leaving certain zinc positions unoccupied [6].

We conclude with a few remarks about the physical properties of these compounds. The electrical conductivities of Rh₂Zn₁₁ and RhZn₁₃ indicate metallic behavior and this can also be expected for RhZn. However, only the latter shows the expected Pauli paramagnetism. For diamagnetic Rh₂Zn₁₁ and RhZn₁₃ apparently the Pauli paramagnetism is so small that it is overcompensated by the core diamagnetism. Thus, we expect for both compounds a very small density of states at the Fermi level. Especially for Rh₂Zn₁₁, which is even more diamagnetic than RhZn₁₃ (Figure 2), this is reflected by the relatively low electrical conductivity. Diamagnetic behavior has also been observed for other intermetallics with γ -brass type structure [43].

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